

EXP 3

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code :

```
def predict(tree, sample):

    if not isinstance(tree, dict):
        return tree

    root = next(iter(tree))

    value = sample[root]

    if value in tree[root]:
        return predict(tree[root][value], sample)
    else:
        return "Unknown"

attributes = ["CGPA", "Aptitude", "Communication"]

tree = id3(data, attributes)

print("Decision Tree:")
print(tree)

new_sample = {
    "CGPA": "High",
```

```

    "Aptitude": "High",
    "Communication": "Medium"
}

result = predict(tree, new_sample)

print("\nNew Sample:")
print(new_sample)

print("\nPredicted Class:", result)

```

output



```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/exp 3.py
Decision Tree:
{'CGPA': {'Low': 'Not Placed', 'Medium': 'Not Placed', 'High': 'Placed'}}

New Sample:
{'CGPA': 'High', 'Aptitude': 'High', 'Communication': 'Medium'}

Predicted Class: Placed
>>>

```